

BusinessLens

Sales & Business Performance Analytics

Python • Excel • Power BI

1. Executive Summary

BusinessLens is an end-to-end analysis of sales transactions covering September 2013 to December 2014. It examines how sales, profit, products, customer segments, countries and discounting relate to one another. The central finding is that revenue alone does not explain business performance.

118.73M Total Sales	16.89M Total Profit	1.13M Units Sold	14.23% Profit Margin
-------------------------------	-------------------------------	----------------------------	--------------------------------

- **Enterprise loses money on high sales:** 19.61M in sales but –0.61M profit, the only loss-making segment.
- **Government is the profit engine:** 11.39M profit, and the top segment in every country.
- **Discounts trade margin for volume:** margin falls from 17.87% (Low) to 9.07% (High discount).
- **Scale and efficiency differ by product:** Paseo leads on sales, Amarilla on margin.

2. Objective and Dataset

The objective was to find where revenue turns into profit and where it does not, across time, products, segments, countries and discount levels. The dataset holds sales transactions with segment, country, product, discount band, units sold, manufacturing and sale price, gross sales, discounts, sales, profit and date.

Data coverage: the data starts in September 2013 and runs to December 2014, so 2013 is a partial year (Sep–Dec). Year-over-year comparisons are therefore not like-for-like.

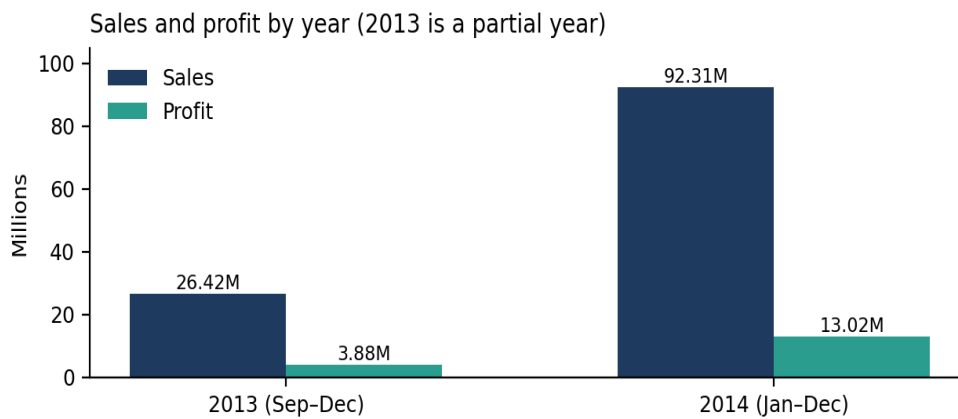
3. Data Preparation and Method

- Inspected structure, data types, missing values and inconsistent entries in Python (Pandas, Jupyter).
- Converted the Date column to a proper date format and validated numeric and categorical fields.
- Set missing Discount Band values to “None” only where no discount was applied, preserving business meaning.
- Exported the cleaned dataset, then analysed it in Excel (pivot analysis) and built a three-page Power BI dashboard (Overview, Product Analysis, Business Analysis).

4. Key Findings

4.1 Sales and profit over time

Sales rose from 26.42M in 2013 (Sep–Dec) to 92.31M in 2014, and profit from 3.88M to 13.02M. 2014 accounts for about 78% of sales and 77% of profit, though part of this gap reflects 2013 being a partial year.



By month, October (21.67M), December (17.37M), November (12.65M) and September (10.88M) were the strongest. Monthly figures are totals across both years, and Sep–Dec appear in both, so the year-end pattern is indicative rather than exact.

So what: Plan inventory and sales capacity around a strong year-end demand period.

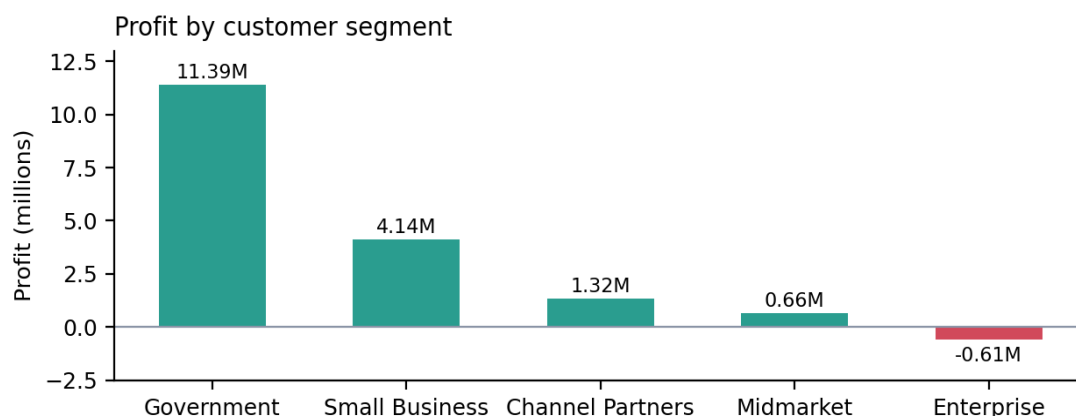
4.2 Products

Product	Sales	Profit margin
Paseo (scale leader)	33.01M	14.53%
Amarilla (margin leader)	17.75M	15.86%
Velo (weakest margin)	18.25M	12.64%

Paseo delivers the highest sales (33.01M) and profit (4.80M), but Amarilla earns the highest margin. Velo sells 18.25M yet has the lowest margin.

So what: Judge products on both contribution and efficiency, not sales volume alone.

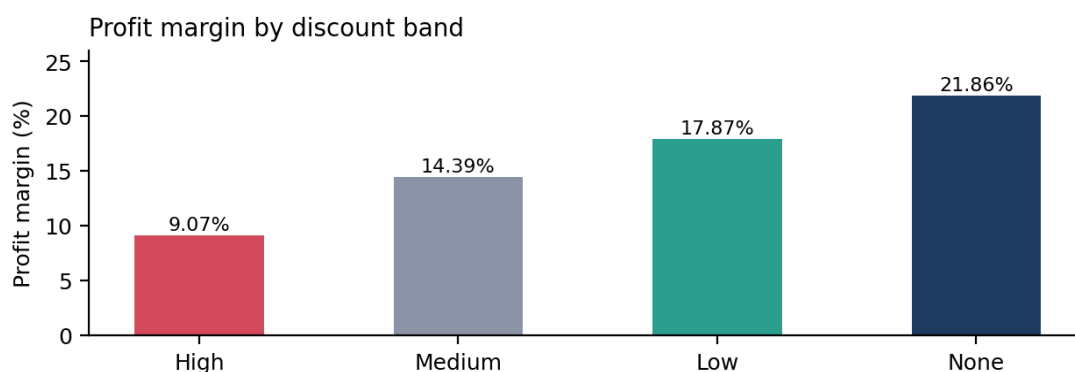
4.3 Customer segments and countries



Government produced 11.39M profit (about 67% of the total) and was the top profit segment in every country: France 2.71M, Germany 2.68M, Canada 2.26M, Mexico 2.04M and the United States 1.70M. Country margins were about 16% for Germany and France, 14% for Canada and Mexico, and 12% for the United States.

So what: Enterprise needs an economics review; segment mix explains country performance better than country totals do.

4.4 Discounting and profitability



The High-discount band generated more sales (37.37M) than the Low band (34.63M), but at roughly half the margin (9.07% vs 17.87%). Because discounts may coincide with particular products or segments, this shows a strong association, not proven cause.

So what: Evaluate discounts by the profit they create, not the sales they generate.

5. Recommendations

Recommendation	Supporting evidence
Review Enterprise economics (pricing, discounts, product mix, cost-to-serve) before pursuing more volume	19.61M sales but -0.61M profit
Make discounting margin-aware; judge promotions by incremental profit	High-discount margin 9.07% vs 17.87% for Low
Protect Paseo while growing higher-margin products such as Amarilla	Paseo 33.01M sales; Amarilla 15.86% margin
Study what drives Government business and replicate it	Top profit segment in all five countries (11.39M)
Plan inventory, capacity and sales resources for Sep–Dec demand	October 21.67M and December 17.37M sales
Track Sales + Profit + Margin + Discount Rate together	Sales alone hid the Enterprise loss

6. Limitations

- 2013 covers only September to December, so growth and seasonality comparisons are indicative.
- The dataset has no cost-to-serve or customer acquisition data, so the causes of Enterprise’s loss cannot be confirmed.
- The discount–margin link is an association and may reflect product or segment mix.
- Only two years of data are available.

7. Conclusion

BusinessLens shows that sales performance and business performance are not the same thing. The strongest signals are Enterprise’s profitability gap, Government’s consistent contribution, product-level margin differences and the link between heavy discounting and lower margins. Combining Python, Excel and Power BI, the project moves from data preparation to analysis, visualisation and actionable recommendations.